Quantification: 535_z27c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	16
interior lacuna count	14
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3680.71	2691.5	2991.77	px^2
major_axis_length_px	lacuna shape	100.666	102.126	39.5915	px
minor_axis_length_px	lacuna shape	50.0371	44.1951	22.0831	px
aspect_ratio	lacuna shape	2.1171	2.1684	0.7132	unitless
eccentricity	lacuna shape	0.8354	0.8873	0.1226	unitless
solidity	lacuna shape	0.7992	0.8223	0.1182	unitless
roots_count	headline	8.5	7.5	4.57	count
ring_length_r30_px	headline	279.714	266	118.6	px
ring_length_r60_px	reported	618.857	539.5	260.695	px
owned_length_px	ownership-dependent	778.079	451.49	628.514	px
edge_count	ownership-dependent	33.7143	18.5	27.8993	count
mean_edge_length_px	ownership-dependent	25.051	24.1272	5.7949	px
perimeter_px	normalised	326.592	283.241	157.523	px
ring_area_r30_px2	normalised	10584	9644	4024.22	px^2
ring_area_r60_px2	normalised	23160.3	20265	8341.28	px^2
in_frame_fraction_r30	normalised	0.9905	1	0.0354	unitless
in_frame_fraction_r60	normalised	0.9829	1	0.0638	unitless
ring_density_r30	normalised	0.0261798	0.0264178	0.00547074	px^-1
ring_density_r60	normalised	0.0265065	0.025848	0.00577052	px^-1
roots_per_100px_perimeter	normalised	2.6753	2.3022	1.059	per 100 px
ring_attached_length_r30_px	network v2	240.429	198.5	119.211	px
ring_attached_length_r60_px	network v2	432.5	324	243.525	px
ring_length_w_r30_px	network v2	316.312	300.229	135.433	px
ring_length_w_r60_px	network v2	700.245	610.857	300.684	px
sholl_crossings_r10	network v2	7.2143	6	3.534	count
sholl_crossings_r20	network v2	8.8571	7.5	4.2762	count
sholl_crossings_r30	network v2	8.9286	7.5	3.8921	count
loop_count	loops, report only	1.4286	0	2.311	count
loop_edge_share	loops, report only	0.0719	0	0.111	share
ring_width_mean_r30_px	width	7.9132	7.8052	0.7874	px

Per field
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measure	value	units
analysed_area_px2	994678	px^2
total_skeleton_length_px	24483	px
canalicular_length_density_per_px	0.024614	px^-1
junction_count	419	count
junction_density_per_px2	0.00042124	px^-2
lacuna_count	16	count

lacunae_per_px2	1.609e-05	px^-2
skeleton_component_count	426	count
mean_component_length_px	57.4718	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0277293	px^-1
field_density_without_flagged_per_px	0.0228844	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1242	px
unattached_component_count	75	count
canalicular_length_density_attached_per_px	0.024614	px^-1
field_length_density_with_unattached_per_px	0.0258626	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	27	count
loop_edge_share	0.0798	share
edge_count	1241	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4552	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	18958	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2942	101.888	45.2188	2.2532	0.8961	0.7542	-1.1126	105.425	685.744	False
2	2441	107.923	42.2715	2.5531	0.9201	0.5582	0.3256	146.253	801.73	False
3	3599	102.364	47.1738	2.1699	0.8875	0.8955	-1.0474	233.848	107.369	False
4	1807	63.8235	38.7415	1.6474	0.7947	0.9372	1.5189	218.143	826.646	False
5	712	35.4872	28.8379	1.2306	0.5828	0.9368	-0.3495	301.131	1012.33	True
6	11051	179.07	107.127	1.6716	0.8013	0.7079	0.7029	439.405	253.828	False
7	5312	120.404	59.815	2.0129	0.8679	0.8459	-1.268	460.68	557.419	False
8	1257	48.9366	34.4065	1.4223	0.7111	0.929	1.2464	562.555	134.776	False
9	2123	56.2494	49.4219	1.1381	0.4775	0.9222	0.3101	574.703	493.107	False
10	2990	94.2448	43.1713	2.183	0.8889	0.8771	0.9473	600.532	860.511	False
11	1556	70.3776	32.2729	2.1807	0.8887	0.7988	1.1709	626.462	984.626	False
12	1310	65.9258	27.032	2.4388	0.9121	0.8501	-1.1711	624.014	302.034	False
13	9402	173.447	80.0495	2.1668	0.8871	0.7022	-0.8455	691.761	675.898	False
14	2231	116.941	28.0885	4.1633	0.9707	0.7914	0.9981	758.602	307.963	False
15	3509	107.726	65.7294	1.6389	0.7923	0.6187	-0.7908	851.443	291.978	False
16	1656	75.1349	31.8764	2.3571	0.9055	0.792	-0.7612	861.306	996.914	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	409	878	1160.69	71	16.3478	8.4707

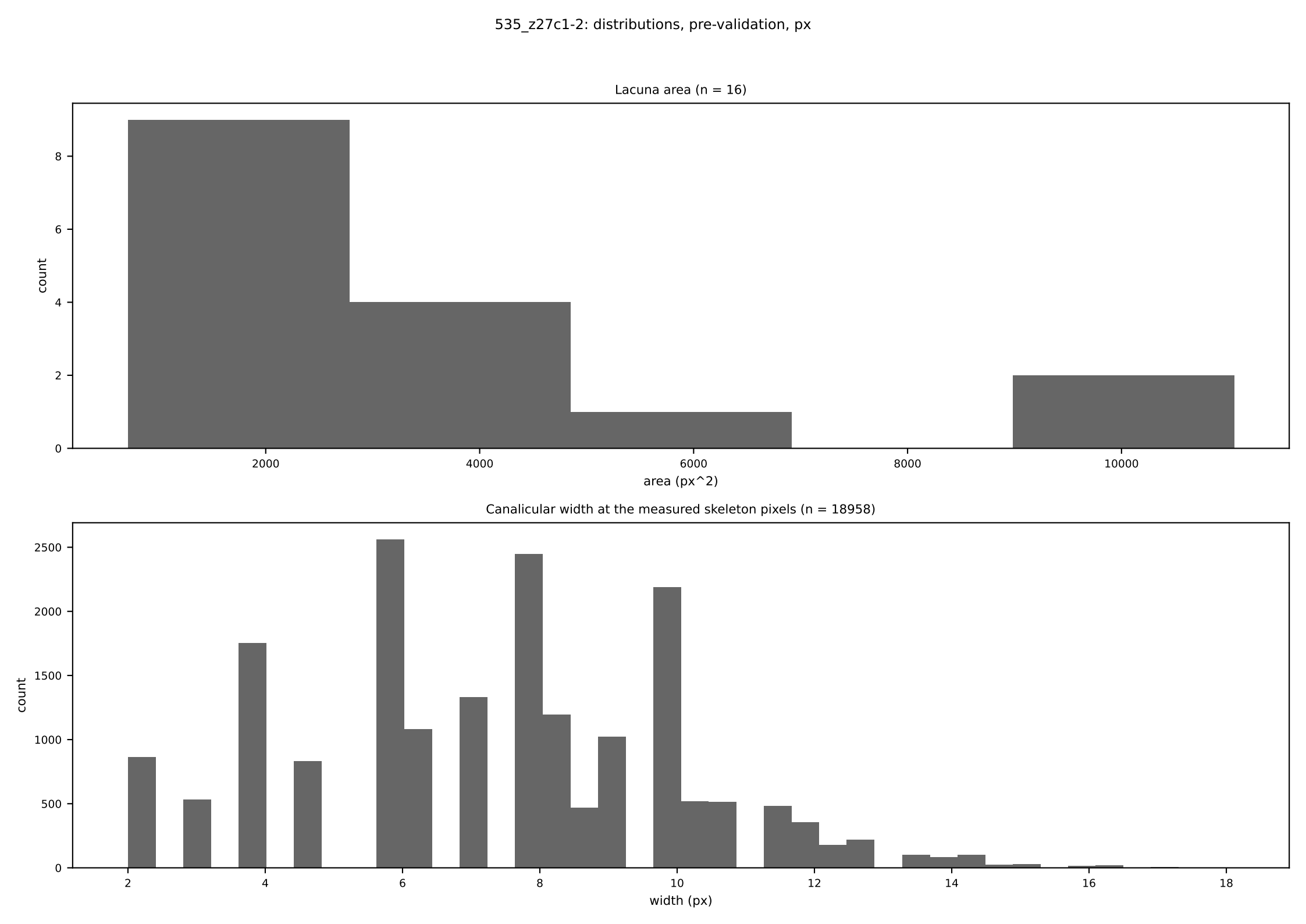
2	12	325	557	377.978	20	18.8989	7.8097
3	6	284	834	1790.48	84	21.3152	7.1459
4	4	140	363	101.154	4	25.2886	7.8006
5	2	92	276	116.841	7	16.6915	7.7797
6	15	445	1043	1159.97	51	22.7446	7.7299
7	18	336	754	935.754	48	19.4949	8.3125
8	7	161	457	525.002	17	30.8825	8.5498
9	5	180	387	295.794	9	32.866	8.4103
10	9	248	522	252.622	11	22.9657	6.7999
11	5	186	402	361.877	17	21.2869	9.9005
12	3	142	269	147.154	4	36.7886	7.8455
13	13	485	1047	1852.93	68	27.249	7.5889
14	4	190	407	331.35	13	25.4885	7.2625
15	10	385	744	1600.34	55	29.0971	7.1584
16	5	125	247	272.066	9	30.2296	7.6324

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	369.203	10657	23260	1	1	0.0383785	0.0377472	2.1668
2	404.546	9790	17493	1	1	0.0331971	0.0318413	2.9663
3	281.948	10538	26647	1	1	0.0269501	0.0312981	2.128
4	184.853	6087	14824	1	1	0.0229998	0.0244873	2.1639
5	112.426	3564	9950	0.5858	0.5585	0.0258137	0.0277387	1.7789
6	645.151	19717	40885	1	1	0.0225694	0.0255106	2.325
7	368.818	12630	27408	1	1	0.0266033	0.0275102	4.8805
8	147.054	6972	18793	1	1	0.0230924	0.0243176	4.7602
9	184.267	7991	18400	1	1	0.0225253	0.0210326	2.7135
10	250.392	9454	20849	1	1	0.0262323	0.0250372	3.5944
11	219.35	6971	13409	0.8676	0.7611	0.026682	0.0299799	2.2795
12	195.48	7849	18629	1	1	0.0180915	0.0144399	1.5347
13	634.625	18182	39984	1	1	0.0266747	0.0261855	2.0485
14	284.534	9498	19681	1	1	0.0200042	0.0206798	1.4058
15	402.073	11840	23982	1	1	0.0325169	0.0310233	2.4871
16	212.551	6244	14852	0.731	0.6597	0.0200192	0.0166308	2.3524

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	383	679	464.647	990.436	10	16	14
2	251	317	367.392	628.286	9	11	7
3	224	543	320.108	931.725	6	8	11
4	90	99	161.853	406.576	3	4	6
5	85	107	98.9411	303.622	3	3	3
6	381	718	507.902	1193.57	14	15	15
7	298	631	369.907	837.683	9	10	9
8	157	331	186.823	522.002	5	5	6
9	139	231	199.61	432.433	4	5	8
10	170	264	280.35	593.428	6	7	6
11	173	288	211.581	451.576	5	7	5
12	107	184	157.297	299.279	4	5	4
13	482	911	555.487	1216.76	14	16	14
14	169	254	210.539	456.848	5	5	6
15	342	605	434.877	842.826	7	10	14
16	107	198	136.983	277.622	6	4	3


535_z27c1-2: distributions, pre-validation, px
Lacuna area (n = 16)
count
area (px^2)
0
2
4
6
8
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 18958)
count
width (px)
0
500
1000
1500
2000
2500
2
4
6
8
10
12
14
16
18